WORKED SOLUTION

STEP III 1992, Question 7

Pure • Vector Geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q7

The key idea

Everything in the question is built from parallelograms and points dividing sides in given ratios, which is exactly what position vectors handle well. Put the origin at A and write every point in terms of the two side vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$. Then find where two of the lines meet, and check that this point lies on the third line.

Solution

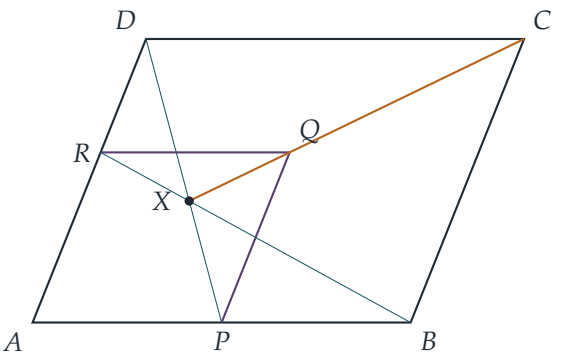
Take A as the origin and let $\mathbf{b} = \overrightarrow{AB}$, $\mathbf{d} = \overrightarrow{AD}$. These are not parallel, because $ABCD$ is a genuine parallelogram. So every point of the plane is $\lambda\mathbf{b} + \mu\mathbf{d}$ for exactly one pair of real numbers (λ, μ), and we can compare coefficients.

Since $ABCD$ is a parallelogram, C has position vector $\mathbf{b} + \mathbf{d}$. Because P lies on side AB and R on side AD, we can write

$$\mathbf{p} = p\,\mathbf{b}, \qquad \mathbf{r} = r\,\mathbf{d}, \qquad 0 \leq p, r \leq 1.$$

Since $APQR$ is a parallelogram, $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{AR}$, so

$$\mathbf{q} = p\,\mathbf{b} + r\,\mathbf{d}.$$



Where BR meets DP

A general point of line BR is

$$\mathbf{b} + s(r\mathbf{d} - \mathbf{b}) = (1 - s)\,\mathbf{b} + sr\,\mathbf{d},$$

and a general point of line DP is

$$\mathbf{d} + t(p\mathbf{b} - \mathbf{d}) = tp\,\mathbf{b} + (1 - t)\,\mathbf{d}.$$

At a common point the coefficients of $\mathbf{b}$ and of $\mathbf{d}$ must agree:

$$1 - s = tp, \qquad sr = 1 - t.$$

Substituting $s = 1 - tp$ into the second equation gives $r - rpt = 1 - t$, so $t(1 - pr) = 1 - r$.

Ask yourself Can we divide by $1 - pr$?

Since $0 \le p, r \le 1$, we have $pr = 1$ only when $p = r = 1$, that is, when $P = B$ and $R = D$. Then $Q = C$, the "line" CQ is not defined, and there is nothing to prove. So we exclude that case, and then $1 - pr > 0$. This also shows BR and DP are not parallel, so they meet at exactly one point. Then

$$t = \frac{1 - r}{1 - pr}, \qquad s = 1 - tp = \frac{1 - pr - p + pr}{1 - pr} = \frac{1 - p}{1 - pr},$$

and the intersection point X is

$$\mathbf{x} = tp\,\mathbf{b} + sr\,\mathbf{d} = \frac{p(1 - r)\,\mathbf{b} + r(1 - p)\,\mathbf{d}}{1 - pr}.$$

X lies on CQ

A general point of line CQ is

$$\mathbf{c} + u(\mathbf{q} - \mathbf{c}) = (\mathbf{b} + \mathbf{d}) + u((p - 1)\mathbf{b} + (r - 1)\mathbf{d}) = (1 - u(1 - p))\mathbf{b} + (1 - u(1 - r))\mathbf{d}.$$

Try $u = \dfrac{1}{1 - pr}$. The coefficient of $\mathbf{b}$ becomes

$$1 - \frac{1 - p}{1 - pr} = \frac{1 - pr - 1 + p}{1 - pr} = \frac{p(1 - r)}{1 - pr},$$

and the coefficient of $\mathbf{d}$ becomes

$$1 - \frac{1 - r}{1 - pr} = \frac{1 - pr - 1 + r}{1 - pr} = \frac{r(1 - p)}{1 - pr}.$$

These are exactly the coefficients of $\mathbf{x}$. So X lies on CQ as well, and the three lines BR, CQ and DP all pass through X. They meet in a point, as required. ■

(The symmetry is reassuring: swapping the roles of B, P, p with D, R, r swaps BR with DP, leaves CQ alone, and leaves the formula for $\mathbf{x}$ unchanged.)

Checks

- If $P = A$ ($p = 0$), then $Q = R$, and $\mathbf{x} = r\mathbf{d}$, which is R. Indeed DP is line DA, which meets BR at R, and CQ is CR, which passes through R.
- If P and R are the midpoints of AB and AD, then $\mathbf{x} = \frac{1}{3}(\mathbf{b} + \mathbf{d})$, one third of the way from A to C. This is the centroid of triangle ABD, as it should be, because BR and DP are two of its medians.
- We checked the concurrency exactly, with fractions, on 5000 random parallelograms and random positions of P and R. Every case matched the formula for $\mathbf{x}$.

Common slips

- Forgetting that the two coefficients can be compared only because $\mathbf{b}$ and $\mathbf{d}$ are not parallel.
- Using the same parameter for two different lines. Each line needs its own (s, t, u).
- Not mentioning the degenerate case $P = B$, $R = D$, where $Q = C$ and the line CQ isn't defined.

Beyond STEP

Notice that the solution never used lengths or angles, only parallel lines and ratios along lines. So the result belongs to *affine geometry*: an affine map (a linear map followed by a translation) sends any parallelogram to a square while keeping concurrency, so it would be enough to prove the result for a square. Questions about lines meeting are also natural in projective geometry, where theorems such as Ceva's and Desargues' live. For a broad, readable tour of these ideas, try Coxeter, *Introduction to Geometry*, or Brannan, Esplen and Gray, *Geometry*.

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